

Exact Convex Coding and Hyperbolic Ledgers for the Equilateral Three-Disk Open Billiard

Route-A source-local certificate HCS-C294

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Abstract

Three equal circular obstacles of radius r have centers at an equilateral triangle of side d . In the sharp strict no-eclipse chamber $d > 4r/\sqrt{3}$, we prove that cyclically reduced cyclic classes are in bijection with periodic-ray iterates—primitive oriented geometric rays paired with positive traversal multiplicities. Primitive classes give unique non-grazing, isolated, hyperbolic primitive rays. We derive the all-period primitive ledger, time reversal, length bounds, and positive determinant-one optical monodromy. We also give a sharp boundary atlas and a deterministic evidence certificate, while separating the source-local collision zeta from every target arithmetic or spectral claim.

1 Model, conventions, and No-eclipse geometry

Let $c_0, c_1, c_2 \in \mathbb{R}^2$ be the vertices of an equilateral triangle of side d , and let

$$K_i = \{x \in \mathbb{R}^2 : |x - c_i| \leq r\}, \quad i = 0, 1, 2.$$

The billiard moves at unit speed in the complement of the interiors of the K_i and reflects specularly. We sample immediately after each collision, so the clock is one bounce. Rays are oriented; changing the initial collision only cyclically shifts the itinerary. Time reversal is not quotiented.

A cyclic word $w = (i_0, \dots, i_{n-1})$ is *reduced* when $i_j \neq i_{j+1}$, including $i_{n-1} \neq i_0$. It is primitive when it is not a proper power. Every cyclic class has a unique form $[w] = [u^m]$ with $[u]$ primitive and $m \geq 1$. A *periodic-ray iterate* is a primitive oriented geometric ray paired with its positive traversal multiplicity m . Length-one reduced words do not exist.

Lemma 1 (Sharp capsule gap). *The convex hull of either two disks is disjoint from the third disk exactly when*

$$d > \frac{4r}{\sqrt{3}}.$$

In that chamber the separation is $\sqrt{3}d/2 - 2r > 0$.

Proof. The convex hull of two radius- r disks is the closed radius- r neighborhood of their center segment. The third center has distance $\sqrt{3}d/2$ from that segment. Removing the capsule radius and the third-disk radius leaves $\sqrt{3}d/2 - 2r$, proving both assertions. $\square$

2 Convex variational coding

For a reduced word w , put $X_w = \prod_{j=0}^{n-1} K_{i_j}$ and, with cyclic indices, define

$$\mathcal{L}_w(q_0, \dots, q_{n-1}) = \sum_{j=0}^{n-1} |q_{j+1} - q_j|. \quad (1)$$

Theorem 1 (Complete periodic-ray coding). *Assume $r > 0$ and $d > 4r/\sqrt{3}$. Cyclically reduced cyclic classes are in bijection with periodic-ray iterates. If $[w] = [u^m]$, its image is the m -fold traversal of the unique primitive oriented exterior ray coded by $[u]$. Every such geometric support is non-grazing, isolated, and hyperbolic, and*

$$[w] \longmapsto [\text{rev}(w)]$$

is the time-reversal involution.

Proof. The continuous convex functional (1) attains a minimum on the compact convex set X_w . Fix a minimizing vertex q_j . If it were in the interior of K_{i_j} , then it would minimize $|x - q_{j-1}| + |q_{j+1} - x|$ without a local constraint. Equality in the triangle inequality places every such minimizer on $[q_{j-1}, q_{j+1}]$. That segment lies in the convex hull of one or two disks other than K_{i_j} and is disjoint from K_{i_j} by Lemma 1, a contradiction. Thus all vertices lie on their assigned circles.

Suppose q and q' were distinct minimizers. Convexity makes $(q + q')/2$ another minimizer. At a coordinate where they differ, strict convexity of the disk puts the midpoint in its interior, contradicting the preceding paragraph. The minimizer is therefore unique.

At a minimizing boundary vertex, the tangential part of

$$\frac{q_j - q_{j-1}}{|q_j - q_{j-1}|} - \frac{q_{j+1} - q_j}{|q_{j+1} - q_j|}$$

vanishes. The normal multiplier is nonzero: if it vanished, the two unit vectors would agree and q_j would lie between its neighbors, again contradicting no-eclipse. The incoming and outgoing tangential components therefore agree and their nonzero normal components are opposite. This is precisely non-grazing specular reflection. The supporting half-plane at an endpoint keeps a flight out of its endpoint disks, and Lemma 1 keeps it out of the remaining disk. Hence the minimizing polygon is a physical exterior ray.

Conversely, the reflection law and the exterior orientation give the normal multiplier with the minimizing sign. They are the first-order optimality conditions for the convex functional on X_w , and hence are sufficient for global minimality. Uniqueness identifies every coded iterate with the constructed minimizer. If $w = u^m$, shifting the minimizing tuple by $|u|$ coordinates preserves X_w and $\mathcal{L}_w$; uniqueness forces the shift to fix the tuple, so the polygon is the m -fold traversal of the primitive polygon for u . Conversely, a smaller geometric collision period makes w a proper power. This proves the primitive/iterate statement. For example, [01] and [0101] share one geometric support but have traversal multiplicities one and two. Reversal follows directly from the itinerary. Non-grazing makes the finite itinerary locally constant, while uniqueness rules out a neighboring periodic ray with that itinerary; equivalently, the hyperbolicity proved below also gives isolation. $\square$

Why the proof is not a finite computation. Compactness gives existence for an arbitrary word length. No-eclipse and strict convexity, not an enumeration cutoff, give boundary contact and uniqueness. The converse is included because a stationary reflected polygon would otherwise not automatically be the unique convex minimizer.

3 Monodromy and uniform length control

Let $\ell_j > 0$ be a free-flight length and let ϕ_j be the incidence angle measured from the obstacle normal, so $\cos \phi_j > 0$. In transverse Jacobi coordinates, free flight followed by a dispersing reflection is represented, up to the standard collision-chart conjugacy, by

$$B_j = R(a_j)F(\ell_j) = \begin{pmatrix} 1 & 0 \\ a_j & 1 \end{pmatrix} \begin{pmatrix} 1 & \ell_j \\ 0 & 1 \end{pmatrix} = \begin{pmatrix} 1 & \ell_j \\ a_j & 1 + a_j \ell_j \end{pmatrix}, \quad a_j = \frac{2}{r \cos \phi_j} > 0. \quad (2)$$

Proposition 1 (Dispersing hyperbolicity). *For every periodic ray, the monodromy $M = B_{n-1} \cdots B_0$ has determinant one, positive entries, and $\text{tr } M > 2$. Thus its multipliers are real positive reciprocals Λ, Λ^{-1} with $\Lambda > 1$.*

Proof. Each factor, and hence their product, has determinant one and strictly positive entries. Thus $M_{11}M_{22} = 1 + M_{12}M_{21} > 1$, and the arithmetic–geometric mean inequality gives $\text{tr } M = M_{11} + M_{22} \geq 2\sqrt{M_{11}M_{22}} > 2$. The characteristic polynomial $\lambda^2 - (\text{tr } M)\lambda + 1$ has two distinct positive reciprocal roots, one larger than one. $\square$

Every flight joins two distinct disks whose centers are distance d apart, so the triangle inequalities give

$$d - 2r \leq \ell_j \leq d + 2r, \quad n(d - 2r) \leq L_w \leq n(d + 2r). \quad (3)$$

At $r = 1, d = 3$, the symmetric orbit [01] has two flights of length one and $a = 2$. Its block squared is

$$\begin{pmatrix} 1 & 1 \\ 2 & 3 \end{pmatrix}^2 = \begin{pmatrix} 3 & 4 \\ 8 & 11 \end{pmatrix}, \quad \text{tr } M = 14.$$

The symmetric orbit [012] has flight length $3 - \sqrt{3}$, incidence cosine $\sqrt{3}/2$, and total length $9 - 3\sqrt{3}$; its positive optical product is again hyperbolic.

4 Exact symbolic and primitive ledgers

Let $A = J_3 - I_3$ be the adjacency matrix: $A_{ij} = 1$ exactly when $i \neq j$. Its eigenvalues are $2, -1, -1$. Therefore the number of collision-marked reduced n -bounce return records, including iterates whose primitive periods divide n , is

$$F_n = \text{tr}(A^n) = 2^n + 2(-1)^n. \quad (4)$$

Every exact-period rooted word has a free orbit of size n under cyclic shift. Möbius inversion consequently gives

$$P_n = \sum_{e|n} \mu(e) F_{n/e}, \quad O_n = \frac{P_n}{n}, \quad (5)$$

where O_n counts primitive oriented geometric rays of collision period n . Reversal can fix some cyclic classes, so it is an involution rather than permission to divide all O_n by two.

With bounce count as the only variable, the source collision-code zeta is

$$\zeta_{\text{coll}}(z) = \exp\left(\sum_{n \geq 1} \frac{F_n}{n} z^n\right) = \det(I - zA)^{-1} = \frac{1}{(1 - 2z)(1 + z)^2}. \quad (6)$$

Equations (4)–(6) are exact for every period because Theorem 1 supplies the geometric bridge.

5 Boundary atlas

Parameter regime	Certified conclusion
$d > 4r/\sqrt{3}, r > 0$	full reduced-word theorem
$d = 4r/\sqrt{3}$	capsule touches the third disk; excluded
$2r < d < 4r/\sqrt{3}$	disks disjoint, but full coding not certified
$d = 2r$	neighboring disks touch
$d < 2r$	neighboring obstacles overlap
$r = 0$	point-scatterer singular limit, excluded
nonreduced or grazing code	outside the collision section

At the equality surface a connecting capsule has zero clearance, so the strict segment-exclusion step in Theorem 1 is unavailable. The weaker disjointness condition $d > 2r$ cannot replace no-eclipse.

6 Evidence, Route-A verdict, and scope

The archived evidence contains exact count rows through $n = 16$, direct word enumeration through $n = 10$, the zeta series through degree 16, six strict geometry cases, two symmetric orbits, and 175 rational optical products. An independent checker imports no producer code; a separate SymPy lane checks the determinant, trace, series, chamber, and Möbius identities. Byte replay, hostile repaired-hash mutations, duplicate-key rejection, and six fresh fixed-epoch manuscript builds close the reproducibility contract. These finite checks audit conventions only and do not prove geometric uniqueness.

The frozen Route-A tuple is

(A0_FAIL, A1_PASS_ANALYTIC, A2_FAIL, A3_FAIL, A4_NATURAL_QUANTIZATION).

A1 passes because word powers are analytically identified with traversal multiplicities and primitive classes with primitive rays at all periods. A0 fails because the alphabet has no target arithmetic-local meaning; A2 fails because bounce count is not an arithmetic clock; A3 fails because (6) is not a target completed function. The exterior Dirichlet Laplacian is a natural quantization only, not a Hilbert–Pólya operator. The overall verdict is `ROUTE_A_REJECTED`, and Route B is locked under `NO_BAD_EULER_OR_ROOT_NUMBER`.

No target Euler factor, root number, automorphy, target divisor law, target functional equation, zero correspondence, or target spectral realization is claimed. Nor do we claim literary priority for classical dispersing-billiard or three-disk mechanisms.

Reproducibility statement. All released JSON and YAML have strict schemas and duplicate-key rejection. The evidence producer is deterministic; the checker, symbolic cross-check, replay, mutation suite, PDF build audit, and self-excluding manifest can be rerun from the package root.

AI-use statement. A generative language model assisted with drafting and verification-code scaffolding. The mathematical claims, source boundaries, exact outputs, and release artifacts were independently checked under the recorded scripts; responsibility for the final content remains with the authors.

Source lineage

Sinai’s dispersing-billiard work supplies historical context [1]; Ikawa supplies neighboring several-convex-body and no-eclipse context [2]; Gaspard and Rice are a direct owner in the three-hard-disk scattering neighborhood [3]. These citations do not assert novelty priority, and the proof above is self-contained.

References

- [1] Y. G. Sinai, “Dynamical systems with elastic reflections: ergodic properties of dispersing billiards,” *Russian Mathematical Surveys* 25 (1970). DOI: [10.1070/RM1970v025n02ABEH003794](https://doi.org/10.1070/RM1970v025n02ABEH003794).
- [2] M. Ikawa, “Decay of solutions of the wave equation in the exterior of several convex bodies,” *Annales de l’Institut Fourier* 38 (1988). DOI: [10.5802/aif.1137](https://doi.org/10.5802/aif.1137).
- [3] P. Gaspard and S. A. Rice, “Semiclassical quantization of the scattering from a classically chaotic repeller,” *Journal of Chemical Physics* 90 (1989). DOI: [10.1063/1.456019](https://doi.org/10.1063/1.456019).